



OFDM Project

Group 11

Adrien Maillet González | 326158

Yassin Alnuaimie | 326279

Contents

1	Introduction	1
2	Implementation	2
2.1	Transmitter	2
2.2	Receiver	3
2.3	Channel Estimation and Phase Tracking	4
3	Results	4
3.1	Basic Data	4
3.2	Impact of the Cyclic Prefix on Spectral efficiency	5
3.3	Phase Tracking	7
3.4	Phase Tracking for a moving microphone	9
3.5	Increase the spectral efficiency	10
3.6	Evolution of the number of Sub-carrier used	11
3.7	Real Data Transmission	12
4	Channel Measurements	14
5	Conclusion	16

1 Introduction

In the context of the course *EE-442 Wireless Receivers: Algorithms and Architectures*, we were tasked with implementing an acoustic Orthogonal Frequency Division Multiplexing (OFDM) transmission system. This project explores the design, implementation, and evaluation of an OFDM-based system tailored for acoustic communication, leveraging its advantages in handling frequency-selective channels.

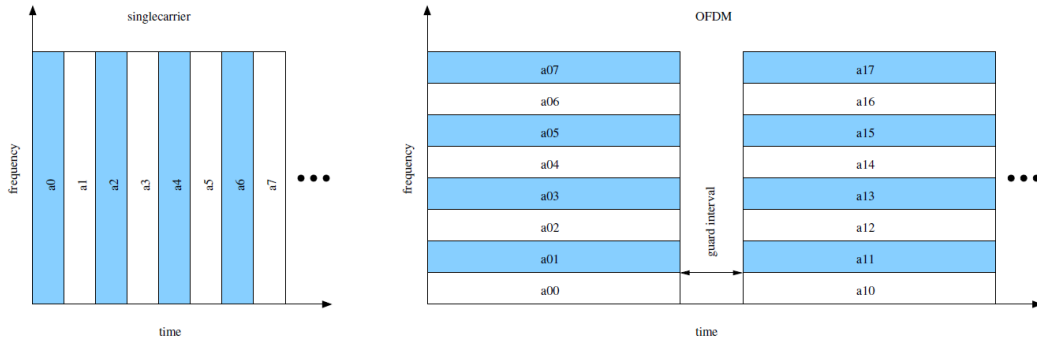


Figure 1: Comparison of single-carrier and multi-carrier (OFDM) transmission

The choice of OFDM over single-carrier modulation and basic multi-carrier modulation is driven by its unique combination of efficiency and robustness. Single-carrier systems transmit data sequentially over a single wide-band channel, as shown in the first part of Figure 1, which leads to significant susceptibility to inter-symbol interference (ISI) in frequency-selective fading environments. Although single-carrier systems benefit from simpler designs in flat channels, their equalization complexity increases substantially when the channel impulse response becomes long compared to the symbol duration.

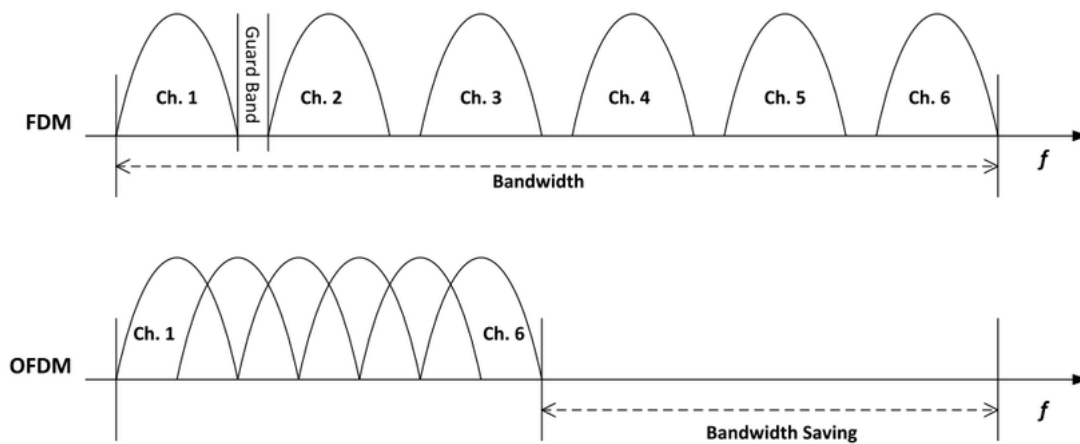


Figure 2: Spectrum Usage: OFDM vs FDM

Basic multi-carrier modulation divides the available bandwidth into multiple sub-channels, introducing guard bands to reduce inter-carrier interference (ICI), as seen in Figure 2 (FDM). While this method mitigates ISI, the guard bands lead to inefficient bandwidth utilization. Additionally, the lack of sub-carrier orthogonality makes channel equalization and interference management more challenging.

OFDM addresses these limitations by ensuring orthogonality among sub-carriers, as illustrated in Figure 2 (OFDM). This approach eliminates the need for guard bands, enabling more efficient use of available bandwidth. Furthermore, the insertion of a cyclic prefix transforms the channel's linear convolution into a circular one, simplifying equalization in the frequency domain. The time-frequency

structure of OFDM, depicted in Figure 1 (OFDM), demonstrates how symbols are distributed across sub-carriers, effectively reducing ISI while maintaining high spectral efficiency.

By implementing an OFDM system, we aim to demonstrate its practical advantages in handling multi-path effects and frequency selectivity, common in acoustic communication scenarios. This report details the system implementation, performance evaluation, and insights gained from exploring its spectral efficiency and error rates under various conditions.

2 Implementation

The implementation of the system can be divided into three main parts: the transmitter, the receiver, and the channel estimation with phase tracking. The transmitter is responsible for preparing the data for transmission, including preamble generation, modulation, and OFDM symbol construction. The receiver processes the incoming signal to recover the transmitted data, handling tasks such as frame synchronization, cyclic prefix removal, and demodulation. Channel estimation and phase tracking serve as the intermediary process that ensures accurate data recovery by compensating for channel effects and phase distortions. These components work together to create a robust and efficient OFDM-based communication system.

2.1 Transmitter

The transmitter was implemented as illustrated in the block diagram (Figure 3). The following steps summarize the implementation:

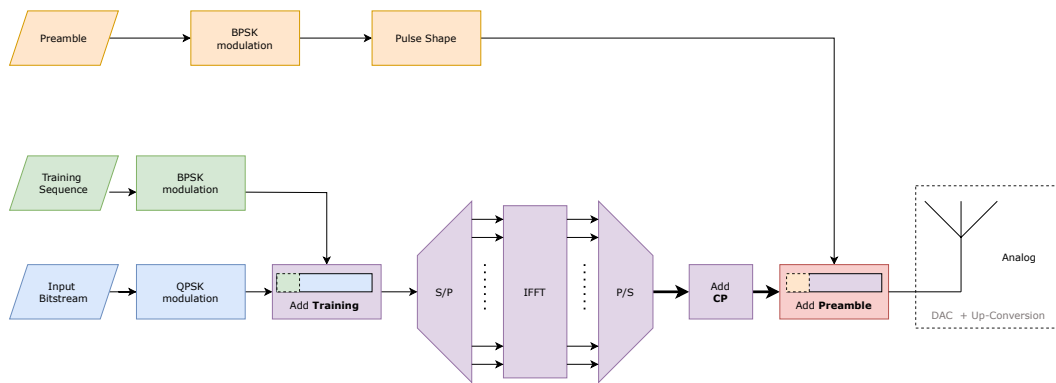


Figure 3: Block diagram of the OFDM transmitter

1. Preamble Generation:

- A BPSK-modulated preamble was generated for synchronization purposes.
- The preamble was up-sampled by a specified oversampling factor.
- The up-sampled preamble was pulse-shaped using a matched filter.

2. Training Sequence Preparation:

- A BPSK training sequence was generated to enable channel estimation.
- The training sequence was periodically inserted into the data stream.

3. Bitstream Mapping:

- Input bits were mapped to QPSK symbols using a predefined constellation.
- The QPSK symbols were arranged into parallel streams suitable for OFDM modulation.

4. OFDM Modulation:

- An over-sampled inverse fast Fourier transform (OSIFFT) was applied to the parallel data streams to generate the OFDM signal.

- A cyclic prefix was prepended to each OFDM symbol to mitigate inter-symbol interference (ISI).

5. Signal Concatenation:

- The preamble, training sequences, and OFDM data symbols were concatenated into a single transmission signal.

6. Normalization:

- The power of the preamble and the OFDM signals was normalized to ensure consistent transmission power. This step prevents the amplification of noise and ensures uniform signal strength throughout the transmission.

7. Up-Conversion:

- The baseband signal was converted to a passband signal by modulating it with a carrier frequency.

2.2 Receiver

The receiver was implemented as illustrated in the block diagram from Figure 4. The following steps summarize the implementation process:

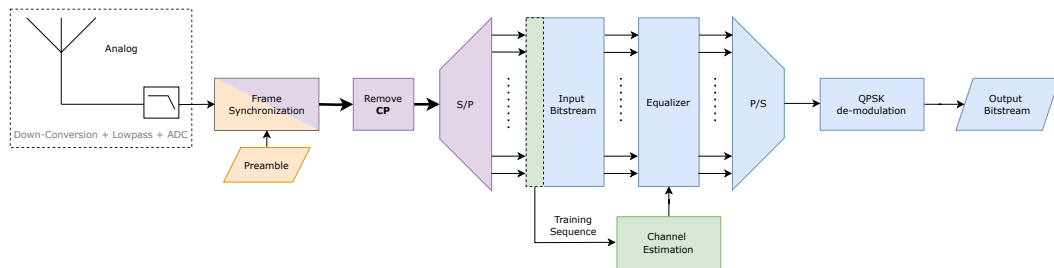


Figure 4: Block diagram of the OFDM receiver

1. Down-Conversion:

- The received passband signal is converted to a baseband complex signal using the carrier frequency.
- A low-pass filter is applied to retain only the desired information around DC.

2. Frame Synchronization:

- A matched filter is applied to the baseband signal for preamble detection.
- The start of the transmitted data is identified using the frame synchronization function.

3. Cyclic Prefix Removal:

- The baseband signal is reshaped into individual OFDM symbols.
- The cyclic prefix is removed from each symbol to extract the useful portion of the signal.

4. FFT Processing:

- A Fast Fourier Transform (FFT) is performed on the OFDM symbols to convert them into the frequency domain.

5. Demodulation and Symbol Mapping:

- The received frequency-domain symbols are mapped to the nearest QPSK constellation points.
- The mapped symbols are decoded into a binary bitstream representing the received data.

2.3 Channel Estimation and Phase Tracking

The channel estimation and phase tracking processes were implemented as shown in the block diagram from Figure 5. The following steps summarize the implementation:

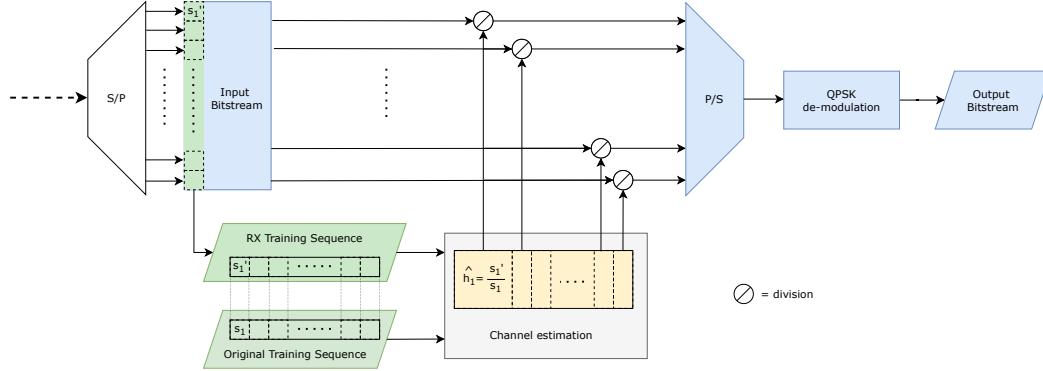


Figure 5: Block diagram of channel estimation

1. Channel Estimation:

- Training sequences embedded in the transmitted signal are used for estimating the channel's frequency response.
- The channel estimation is performed by dividing the received training sequence by the known transmitted training sequence:

$$\hat{h}_i = \frac{s_{i,RX}}{s_{i,TX}}$$

where $s_{i,RX}$ and $s_{i,TX}$ are the received and transmitted training symbols, respectively.

2. Phase Tracking:

- Phase distortions introduced by the channel are estimated and corrected independently for every sub-carrier.
- Two phase tracking methods:
 - Training symbols are periodically inserted (Block-type Training) to measure and correct phase deviations (Discrete Phase Tracking).
 - The Viterbi-Viterbi algorithm is used to estimate and refine phase offsets when enabled (Continuous Phase Tracking).
- The phase correction process updates the received symbols to compensate for distortions:

$$\text{Corrected Data} = \frac{\text{Received Signal}}{\hat{h}_i} = \frac{\text{Received Signal}}{|\hat{h}_i|} \cdot e^{-j\angle\hat{h}_i}$$

where $\angle\hat{h}_i$ represents the estimated phase offset for each sub-carrier.

3 Results

In this section we will study the performance of our system for different types of data transmission and different channel conditions. We will start with a short bitstream and good channel conditions. We will then disturb the channel and analyze the performance evolution of our system. Eventually, we will transmit real data (images) and we will determine how the choice of the parameters influence the quality of our data at the receiving end.

3.1 Basic Data

In this section we will transmit one training symbol and one data symbol using 256 carriers, a spacing of 5Hz and a cyclic prefix of half the OFDM symbol.

We place the microphone at 15cm from the speakers and we obtain the following result.

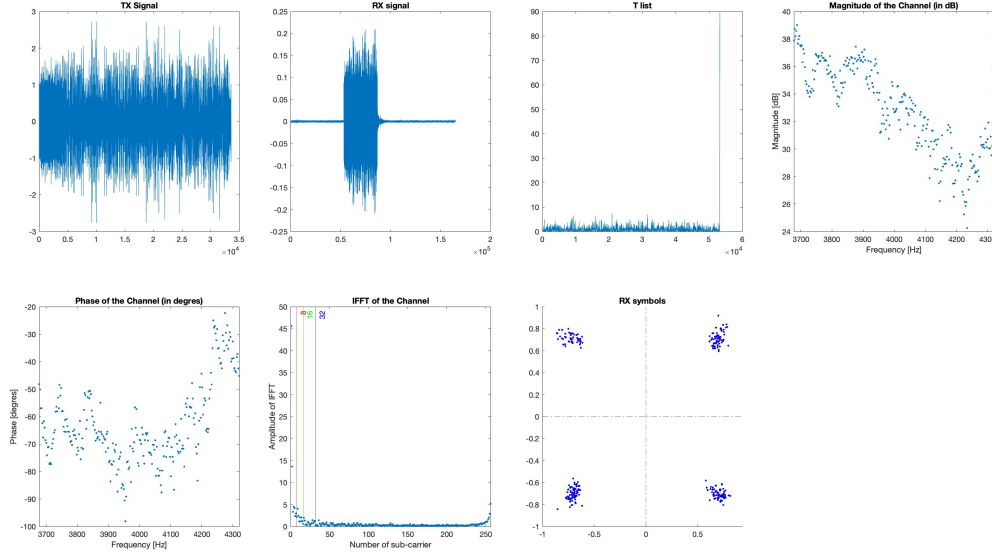


Figure 6: Summary plot of the *Basic Data* transmission

Using Figure 6, we confirm that the acoustic OFDM transmission system works under good channel conditions. The **T list** plot shows us that the frame synchronization happens without any ambiguities and we are able to identify the start of the data. Concerning the constellation (see the **RX Symbols** plot), we observe that all the received symbols are very close to the actual constellation points. The **IFFT of the channel** plot represents the impulse response of the channel. Here we can see that using a cyclic prefix of half the symbol length is very much an overestimate. Instead of using a cyclic prefix of length 128, we could use one of length 32 without degrading the performance too much (we will test this estimation). Finally, using the **Magnitude of the Channel (in dB)** and the **Phase of the Channel (in degrees)** plots, we can study the evolution of the channel depending the carrier frequency. Regarding the magnitude, we observe a decrease in the magnitude when increasing the frequency. This makes sense as higher frequencies are linked with higher losses when propagating in the air. Concerning the phase, we notice a phase shift for higher frequencies, confirming the degradation of performance for higher frequencies.

3.2 Impact of the Cyclic Prefix on Spectral efficiency

We will now compute the spectral efficiency of our system using the current configuration and we will compare this value to the one we obtain using the minimum required length cyclic prefix. The spectral efficiency of an OFDM symbol can be calculated as:

$$\eta = \frac{R_b}{B}$$

With R_b being the total data rate in bits per second, and B being the total bandwidth in Hertz. As we use $N = 256$ carrier frequencies separated by a frequency range of $\Delta f = 5\text{Hz}$, the overall bandwidth is:

$$B = N \cdot \Delta f = 1280\text{Hz}$$

In order to compute the data rate, we need to evaluate first the total number of data bits sent and then compute the total time required to send them. Let n be the number of data symbols sent for the Basic Data scenario and let T_{tot} be the total time taken to send them (we use symbols in the computations as it is easier to recognize the values; 1 data symbol = 2 data bits). Note here that included in the total time, we have the preamble, the CP, the training symbols and the data symbols.

$$\text{len}(\text{data}) = n = 256 \text{ symbols}$$

For the time taken to send these bits, we need some information about the sampling frequency and the oversampling factors of both the preamble and the data. In our system we use: $f_{\text{sampling}} = 48\text{kHz}$, the symbol rate of the data is $f_{\text{data}} = N \cdot \Delta f = 1'280\text{Hz}$ and we pick a symbol rate for the preamble to be $f_{\text{preamble}} = 1'000\text{Hz}$. Therefore, the oversampling factors of the preamble and the data are, respectively: $OS_{\text{preamble}} = \frac{48'000}{1'000} = 48$ and $OS_{\text{data}} = \frac{48'000}{1'280} = 37.5$.

- As the preamble length is taken to be 100 bits, the time taken to send the preamble is:

$$\Delta t_{\text{preamble}} = \frac{\text{len}(\text{preamble}) \cdot OS_{\text{preamble}}}{f_{\text{sampling}}} = 0.1 \text{ seconds}$$

- We send one OFDM symbol with one training symbol. Thus the data length is $\text{len}(\text{data} + \text{training}) = 2 \times n = 512 \text{ symbols}$. Considering first that the CP is half the OFDM symbol length, we have: $\text{len}(\text{data} + \text{training} + \text{CP}) = \text{len}(\text{data} + \text{training}) + \frac{\text{len}(\text{data} + \text{training})}{2} = 768 \text{ symbols}$. The time taken to send the data is:

$$\Delta t_{\text{data}+\text{CP}} = \frac{\text{len}(\text{data} + \text{training} + \text{CP}) \cdot OS_{\text{data}}}{f_{\text{sampling}}} = \frac{768 \cdot 37.5}{48'000} = 0.6 \text{ seconds}$$

Therefore, the spectral efficiency of our system with a cyclic prefix of half the OFDM symbol length is:

$$\eta_{\text{len}(\text{CP})=N/2} = \frac{\frac{256}{0.1+0.6}}{1280\text{Hz}} = 0.286_{[\text{symbol/s/Hz}]} = \mathbf{0.571 [\text{bit/s/Hz}]}$$

If we decide to use now a CP of minimum required length, i.e., $\text{CP}' = 32 \text{ symbols}$; the only difference would be a decrease in the data sending time. Indeed, we will now have: $\text{len}(\text{data} + \text{training} + \text{CP}') = \text{len}(\text{data} + \text{training}) + 32 \times 2 = 576 \text{ symbols}$. Consequently,

$$\Delta t_{\text{data}+\text{CP}'} = \frac{\text{len}(\text{data} + \text{training} + \text{CP}') \cdot OS_{\text{data}}}{f_{\text{sampling}}} = \frac{576 \cdot 37.5}{48'000} = 0.45 \text{ seconds}$$

The spectral efficiency of this system would be:

$$\eta_{\text{len}(\text{CP}')=32} = \frac{\frac{256}{0.1+0.45}}{1280\text{Hz}} = 0.364_{[\text{symbol/s/Hz}]} = \mathbf{0.727 [\text{bit/s/Hz}]}$$

If we reduce the CP to the required length only, i.e., to only 32, we would get a better spectral efficiency of our system. When we test our system for a channel length of 15cm we still get no bit errors for one OFDM symbol. We see here the link between the spectral efficiency and the size of the Cyclic Prefix. In Figure 7, we plotted the evolution of the spectral efficiency in terms of the cyclic prefix length.

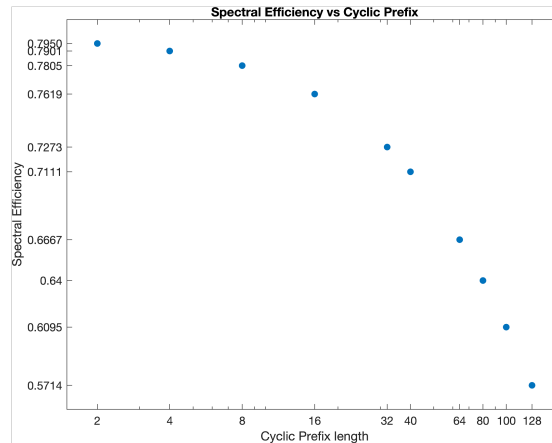


Figure 7: Evolution of the Spectral Efficiency in terms of the length of the Cyclic Prefix

Here we can clearly see the dependence. By reducing the length of the Cyclic Prefix, the gains in terms of Spectral Efficiency do not improve too quickly. Similarly by increasing the length of the Cyclic Prefix, there is a point where the efficiency decreases too fast.

We will now observe experimentally how the error changes over time for a varying CP. The following table summarizes the parameters we used in this simulation.

$f_{carrier}$	4 kHz
Δx (distance microphone to speaker)	2.00 m
N (number of sub carriers)	256
Δf	5 Hz
#OFDM symbols	5

Table 1: Summary of the main simulation parameters

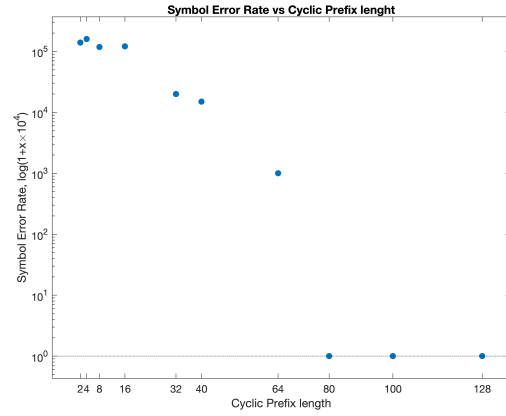


Figure 8: Evolution of the Symbol Error Rate in terms of the cyclic prefix length

These experimental results make a lot of sense. We observe a decreasing error for an increasing Cyclic Prefix in length. Here the value of $length(CP) = 32$, seems to be a good trade-off between spectral efficiency and symbol error. In this section we decided to use the spectral efficiency as the efficiency of the transmission over the available bandwidth. However, we could also use a simpler definition of the efficiency. We could define the efficiency of the transmission as:

$$\varepsilon = \frac{N}{N + length(CP)}$$

In this definition the efficiency is unit-less and the evolution of the error in terms of the Cyclic Prefix length is:

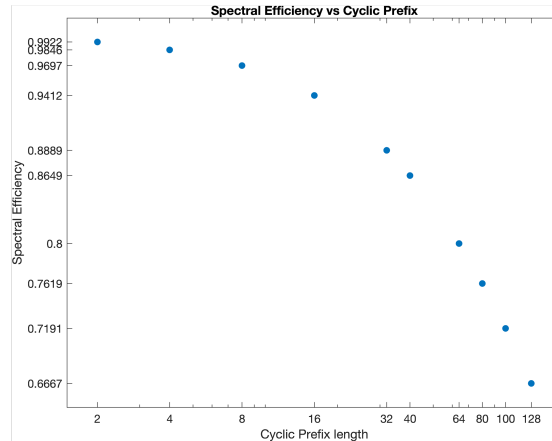


Figure 9: Spectral Efficiency in term of the length of the Cyclic Prefix

In this Figure, we have the exact same evolution of the efficiency as the one found in Figure 7. This efficiency only accounts for the influence of the CP length. It confirms the choice of using a Cyclic Prefix of length 32.

3.3 Phase Tracking

We will now send a longer bitstream in order to evaluate experimentally the maximum number of symbols we can send before requiring a continuous phase tracking. The only parameter we will vary here is the length of the transmitted bit stream. We will still use $N = 256$ carrier frequencies, a spacing of $5Hz$ and a cyclic prefix of length 32. Concerning the physical experimentation, we realized a constant spacing of only 15 cm was not sufficient to observe a significant evolution of the error.

Therefore, we spaced the microphone and the speaker at two meters from each other. The results we have are the following.

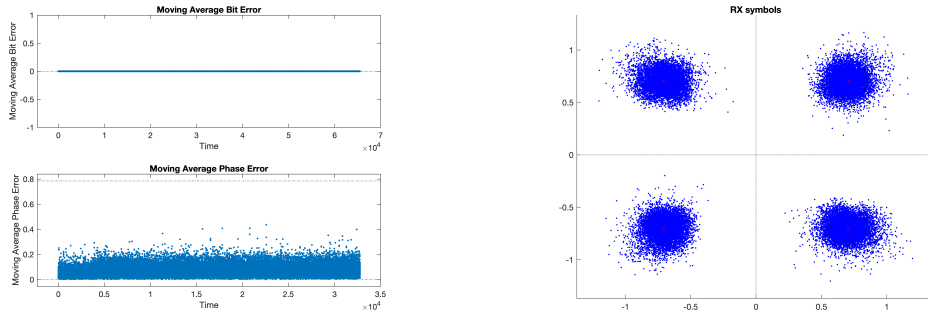


Figure 10: Evolution of the error with time when sending 128 symbols

We observe here that we do not need to implement phase tracking when sending 128 symbols. Following this experiment, we tried to send 512 symbols while keeping both the microphone and the speaker at the same position.

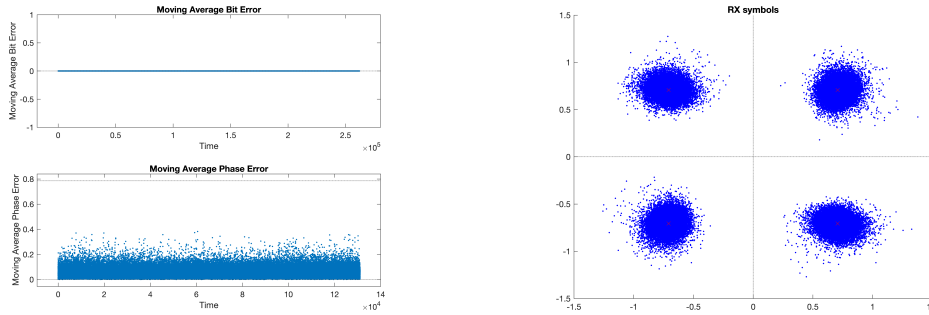


Figure 11: Evolution of the error with time when sending 512 symbols

Thus, if the physical system is static over the simulation period and the room in which the experiment is performed is calm, we can conclude that there is no need to perform continuous phase tracking. One of the reasons behind this good performance even for a very long input bitstream could be the good sampling between the laptop we used for the experiments and the microphone/speaker system. Here as an example we will show how the system performs when using continuous phase tracking. We implemented the Viterbi-Viterbi algorithm and we use as weights of the correction angle 0.5 for the previous angle correction and 0.5 for the angle error found using Viterbi. The results when sending 128 symbols are the following:

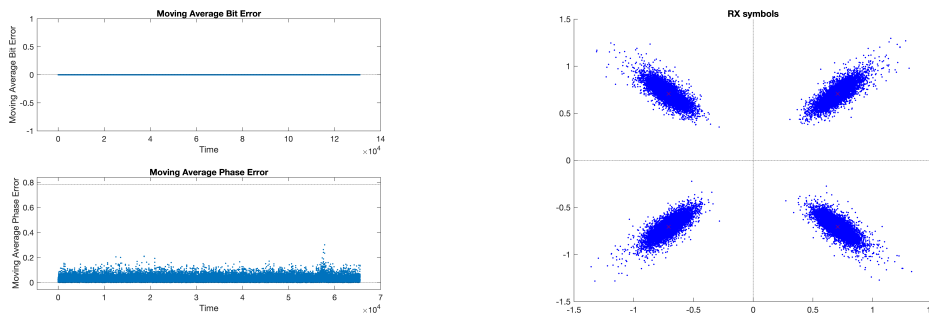


Figure 12: Evolution of the error with time when sending 256 symbols with phase tracking

Thus, using continuous phase tracking will improve the robustness of the system even for a static system. Here we have a phase error around 0.1 radians. The average phase error without phase

tracking was around 0.2 radians. In this scenario phase tracking is not able to improve the accuracy of our transmission as we are already able to accurately recover 100% of our symbols. However, we will see how it will increase the performance for a moving system in the next section.

3.4 Phase Tracking for a moving microphone

Moving the microphone has a direct impact on the phase of our signal. Focusing on the carrier frequency at $f_{carrier} = 4kHz$ and considering the speed of sound in the air to be $v_{sound} = 343m.s^{-1}$, we conclude that the wavelength of the carrier frequency in the air is $\lambda_{f_{carrier}} = 8.575cm$. As we can start having symbol errors when shifting the phase by $\frac{\pi}{2}$, we conclude that moving the microphone only by $2.1cm$ will generate some errors. We start by testing this for our system without phase tracking for 32 symbols.

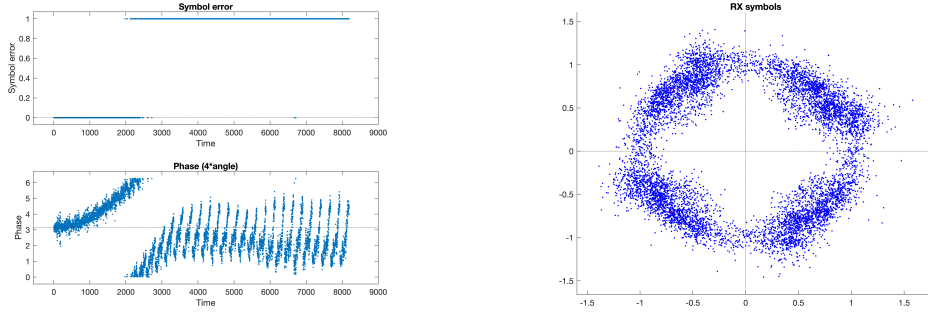


Figure 13: Evolution of the error with time when sending 32 symbols without phase tracking, when the microphone move by a quarter of a wavelength

The results we obtain in Figure 13 are exactly what we expected to find. By moving the microphone to a quarter of a wavelength from the speaker, we shift every symbol to the next in the constellation. As a consequence, the symbol error stays at 1 for the rest of the simulation. Indeed, by moving the microphone by a quarter of a wavelength from the speaker, every symbol will be shifted to the next quadrant. Another very interesting result that we obtain from the left plot is related to the amplitude of the phase in terms of time. We observe a very small amplitude at the beginning, i.e., there are very few errors in the phase at the beginning. As we decrease the distance between the two objects, the phase increases. However, since we have 256 different sub-carriers, the displacement in distance will result in different phase shifts for different sub-carrier frequencies. Here, smaller sub-carrier frequencies, i.e. longer wavelengths, will be less affected than the carrier frequency. They will correspond to the bottom of the phase spread. The exact opposite will happen to the higher sub-carrier frequencies, i.e. shorter wavelengths. Moving the microphone by $2.1cm$, will shift the phase by more than $\frac{\pi}{2}$. As we have 256 carriers and a frequency gap of $\Delta f = 5Hz$, the resulting phase shift for the smallest and the longest wave lengths are:

	$f_{carrier} - \frac{N}{2} \cdot \Delta f$	$f_{carrier}$	$f_{carrier} + \frac{N}{2} \cdot \Delta f$
f (in Hz)	3'360	4'000	4'640
λ (in cm)	10.208	8.575	7.392
Δx (in cm for a phase shift of $\frac{\pi}{2}$)	2.552	2.1	1.848
$\Delta\varphi_{\Delta x=2.1cm}$ (in rad)	$\frac{\pi}{2} - 0.278$	$\frac{\pi}{2}$	$\frac{\pi}{2} + 0.214$

Table 2: Frequency dependent phase shift for a decreasing channel length

We will now observe to what extend the continuous phase tracking is able to correct for this phase shift.

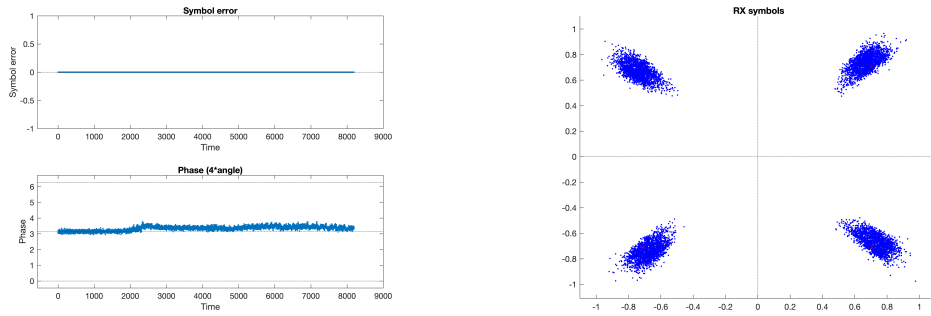


Figure 14: Evolution of the error with time when sending 32 symbols with phase tracking, when the microphone move by a quarter of a wavelength

The issue with this phase tracking is that it assumes the evolution of the phase error is relatively slow. As a consequence, a quick phase shift could be undetected by the continuous phase tracking. In order to correct for this behavior, we decide to add new training symbols inside the data. Instead of only using one training symbol at the start of our OFDM transmission we have the option to add training sequences at a fixed rate inside the data. This enables the system to correct itself from any abrupt phase shift. We will now test the performance of our system with this feature alone and with this feature coupled with the continuous phase tracking. First we will add only one phase tracking in the middle of the sample and then we will add 3 training symbols equally spaced between the signal symbols.

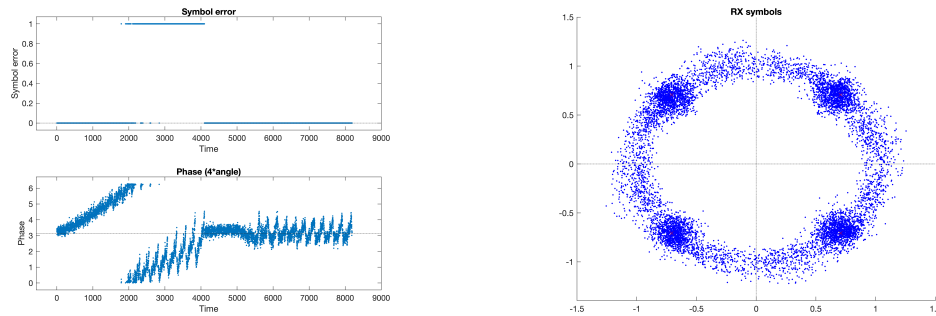


Figure 15: 2 training symbols

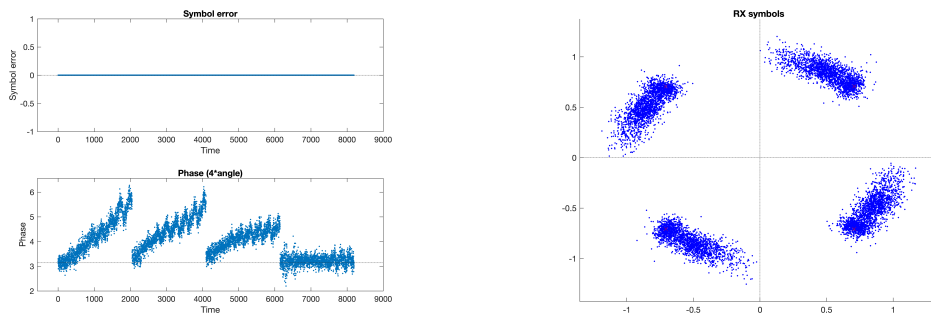


Figure 16: 4 training symbols

3.5 Increase the spectral efficiency

In order to increase the spectral efficiency we can reduce the frequency spacing between carriers. We currently use a spacing of 5Hz . We will gradually change this spacing and observe the consequences on the performance of our system. The following table summarizes the parameters we used in this simulation.

$f_{carrier}$	4 kHz
Δx (distance microphone to speaker)	2.00 m
N (number of sub carriers)	256
$length(CP)$	128
#OFDM symbols	5

Table 3: Summary of the main simulation parameters

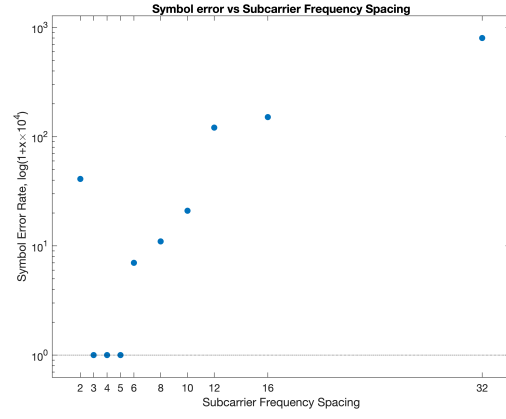


Figure 17: Evolution of the Symbol Error Rate in terms of the sub-carrier frequency spacing

In Figure 17, we observe the experimental results of a varying sub-carrier frequency spacing. We observe here two very interesting results. First, we have that a spacing of $5Hz$, results in no errors. Which confirms it is a good choice in terms of frequency spacing. We also have no errors for the values around $\Delta f = 5Hz$. However, when we decrease too much the frequency spacing we start to observe the apparition of errors. Indeed, there is a limit up to which we can squeeze the sub-carriers. The risk is that the Orthogonality of the technique is not respected. Indeed, we would need to have much sharper sine cardinal functions. Which would result in a wider bandwidth. Eventually, for larger frequency spacings as well, we observe an increasing error. By increasing the frequency spacing, we increase the bandwidth. Thus, for a fixed carrier frequency, we may generate some interference between the higher frequency and the lower frequency sub-carriers. Indeed, taking $\Delta f = 32Hz$, the bandwidth becomes $B = \Delta f \cdot N = 8'192Hz$. As we recall from Table 3, the carrier frequency we used for this experiment was $f_{carrier} = 4'000Hz$. As we have $\frac{B}{2} > f_{carrier}$, there will be spectral folding and the errors would occur for the sub-carriers with the smaller and the highest frequencies. In Figure 18, we observe visually how the highest and lowest frequency are affected.

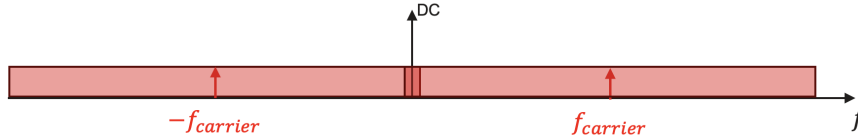


Figure 18: Representation of the Spectral Folding in the frequency domain

3.6 Evolution of the number of Sub-carrier used

Another way we can improve the efficiency of our system is to increase the number of sub-carriers we use. As we will try to send images in the next section, we understand that increasing the number of sub-carriers could drastically reduce the transmission time. However, there is also a limit on the number of sub-carriers we can use before having uncontrollable errors. In the following experiment we double the number of sub-carriers to understand how the error would evolve. The following table summarizes the parameters we used in this simulation.

$f_{carrier}$	4 kHz
Δx (distance microphone to speaker)	2.00 m
Δf	5
$length(CP)$	$\frac{N}{2}$
#OFDM symbols	5

Table 4: Summary of the main simulation parameters

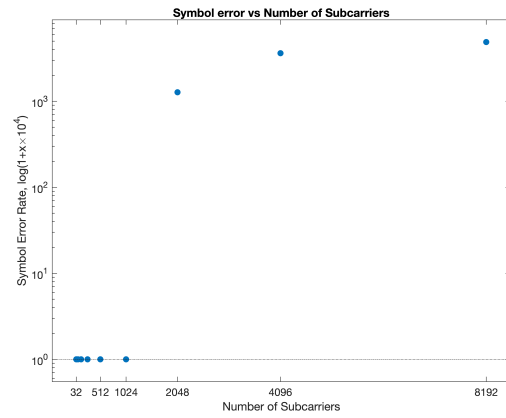


Figure 19: Evolution of the Symbol Error Rate in terms of the number of sub-carriers

By doubling the number of sub-carriers, we halved the data transmission time. However, the bandwidth would also double. As we use a relatively high frequency spacing (refer to Table 4), our bandwidth is very wide. Here, the spectral folding error would occur only for a number of sub-carriers of 8'192. However, using more than 1'024 sub-carriers already results in errors. A solution we could use to mitigate this error would be to use a higher carrier frequency and to reduce the frequency spacing. If we still have orthogonality, we would drastically reduce the transmission time.

3.7 Real Data Transmission

Sending images via our OFDM system combines all the difficulties we previously encountered and adds the problem of having a high PAPR. A high PAPR is a critical issue for an OFDM-based transmission system. It could lead to significant distortion and could result in amplitude clipping. Additionally, as the number of sub-carrier increase, the PAPR influence is exacerbated. When sending images, we most likely have areas with similar colors. And if these colors are associated with a high energy, they would result in a peak of the energy. Considering now, an image with multiple color levels, the average energy will not be high. Thus as we will normalize the signal before sending it, we might have to make sure the parts of the signal with the lowest energy will not be crushed by the rest.

The image we will try to send is the following:



Figure 20: Original Image

In a first stage, we tried to send the image with a static system using 1'024 sub-carriers. the image was sent without any problem, confirming the implementation was functional.

pyramid.png



Figure 21: Reconstructed received image

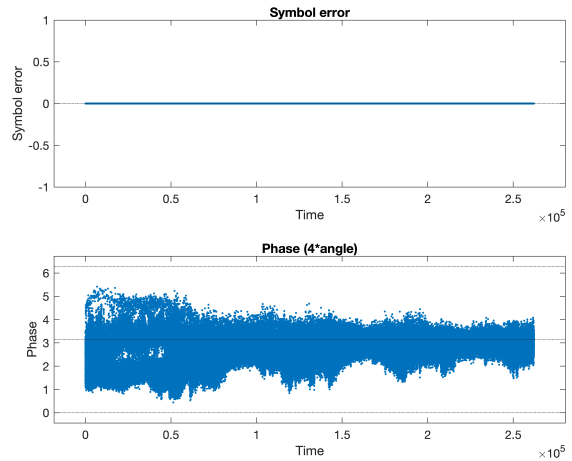


Figure 22: Evolution of the Symbol Error and the phase over the transmission time

In order to test the robustness of the image transmission, we increased the distance between the microphone and the speaker during transmission. Here are the results without phase tracking (Figure 23) and only with continuous phase tracking (Figure 24) respectively.

pyramid.png

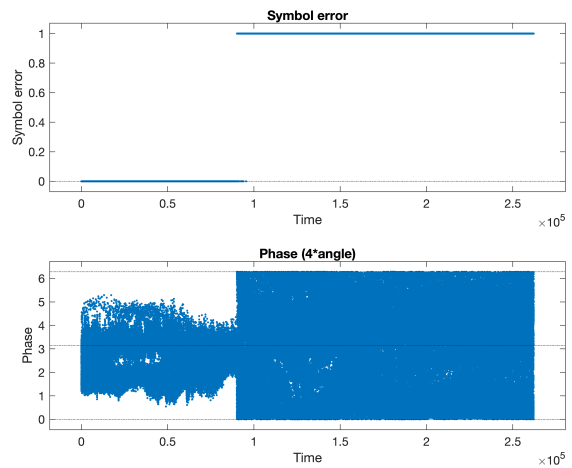
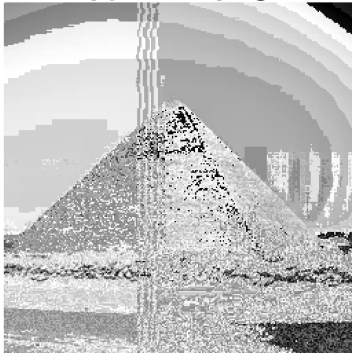


Figure 23: No phase tracking moving device

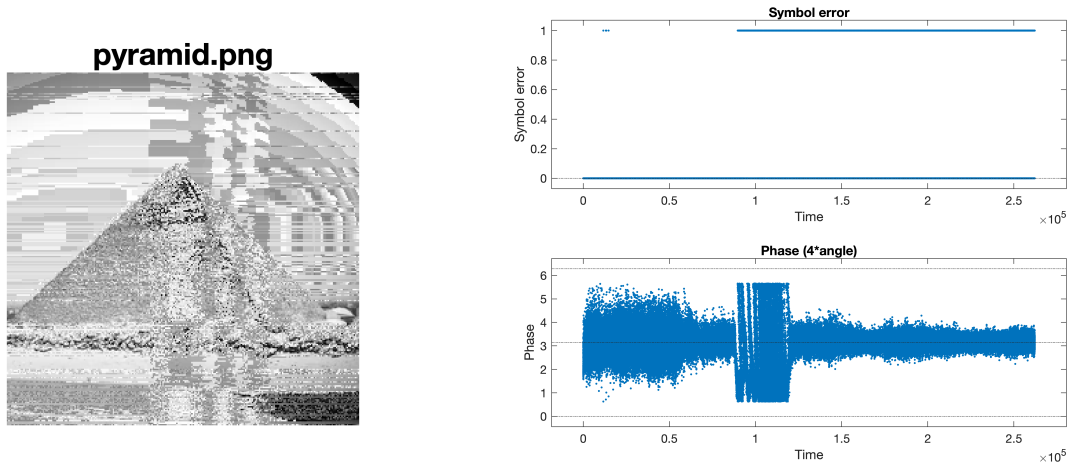


Figure 24: Continuous Phase tracking for a moving device

These two results showcase the poor performance of our image transmission when no phase tracking of a simple phase tracking method only. We previously tested the continuous phase tracking for a simple data transmission and we observed a good performance (see Figure 14). Here however, the phase tracking is not able to correct for phase errors in real time. Therefore, as we increase the distance and we introduce the first symbol errors, we observe a consistent number of symbol errors. Eventually, we tried to use both continuous tracking and multiple training. This solution is the most robust to channel variation and high PAPR distortions. The results we obtain in this case are the following:

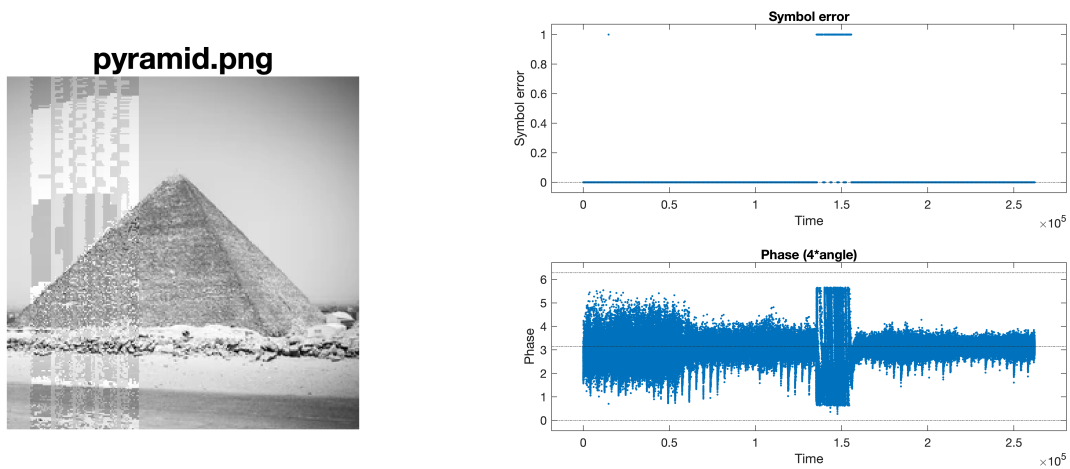


Figure 25: Continuous Phase tracking and 64 Training Sequences for a moving device

When both tracking methods are implemented in parallel, the performance of our system is greatly improved. We do observe some errors which correspond to the symbols sent during the movement of the speaker. However, as we can see on the symbol error plot. Even if we do have some symbol errors around the time 1.5×10^5 , we are able to recover from them and go back to an accurate data transmission. To have proper data transmission, in practice we would always combine both of these correction methods. As mentioned in class, comb-training could be a robust option against the channel variations.

4 Channel Measurements

Finally, we will analyze the channel. How different channels could affect the performance of the transmission and how would the channel react to different stimuli during the transmission. First we will estimate the channel for a static environment, and then we will both decrease and increase the

distance between the speaker and the microphone and finally, we will introduce an obstacle during the transmission. For this series of observations, we used a short channel of approximately 15cm (if not specified otherwise) and we use a carrier frequency of $f_{\text{carrier}} = 14\text{kHz}$.

The colors used in these figures represent the channel magnitude and phase at different instants of the transmission. Every color corresponds to one training sequence and the order in which training sequences are sent is the following:



Figure 26: Color associated with every training sequence

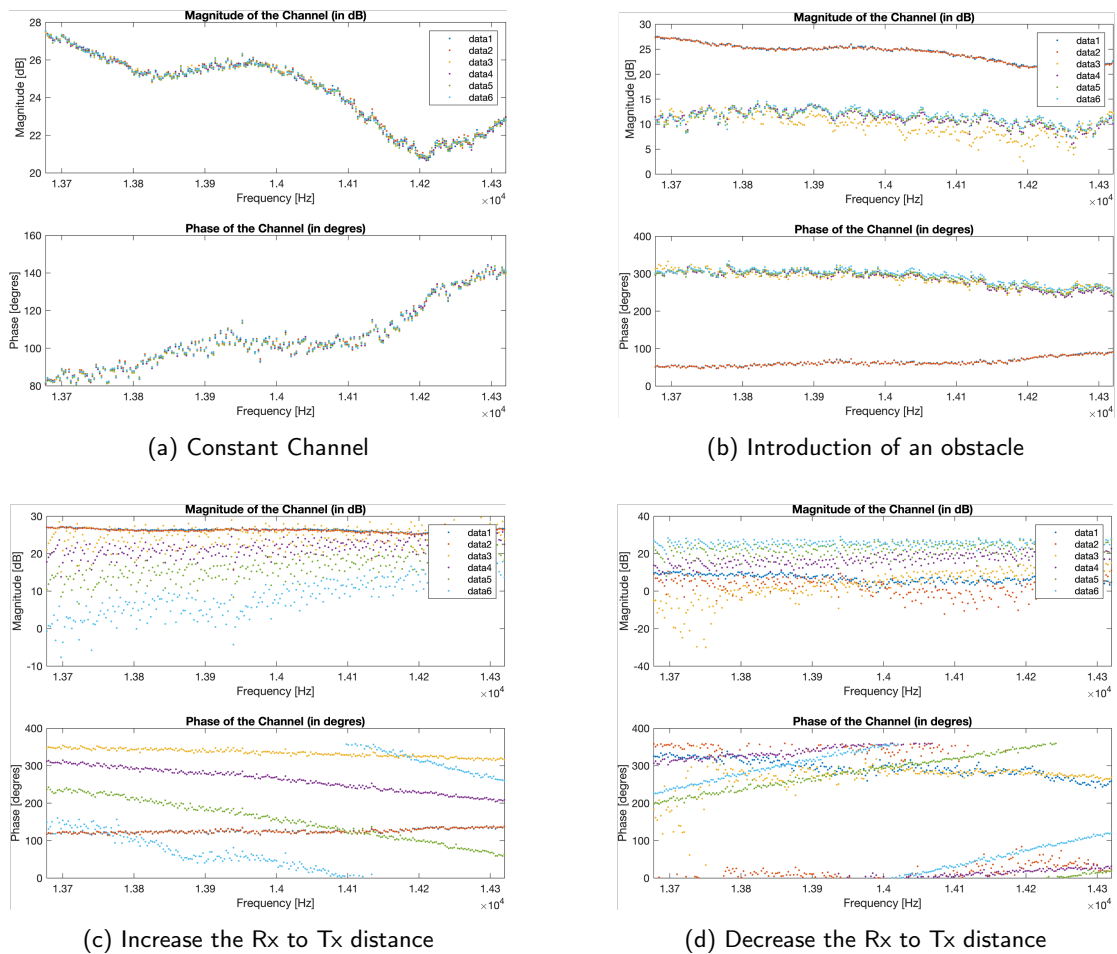


Figure 27: Evolution of the channel over time for different channel conditions

- We observe in Figure 27a that the channel stays very consistent over time. The magnitude of the channel is relatively high, around 24dB and the phase does not vary a lot between each sub-carrier.
- When we suddenly place an object between the receiver and the transmitter, we have a sudden drop in magnitude and a sudden increase in the phase. As there is no direct path, the magnitude will decrease (-15dB). Concerning the phase we introduce a phase shift of 210 degrees. If we do not introduce a training sequence after this phase shift, there will be no way for us to recover the correct data and we will have a constant error.

- As we change the distance separating the microphone and the speaker, we modify the magnitude of the channel. As we can see in Figure 27c, by increasing the size, we decrease the magnitude. Alternatively, by decreasing the distance, we increase the magnitude and we improve the SNR (see Figure 27d). The phase evolution however is similar for both scenarios. We have a relatively flat phase distribution. As we vary the distance, the slope of the phases becomes steeper. This is due to the difference in wavelength for the different sub-carrier frequencies. A smaller frequency, will have a longer wavelength and would thus be less affected by a change in distance. The opposite is also true.

The conclusions we can draw from these plots is that the combination of the continuous and the discrete phase tracking could fix all the errors introduced by a varying channel.

5 Conclusion

In this project, we successfully implemented an acoustic Orthogonal Frequency Division Multiplexing (OFDM) transmission system as part of the EE-442 Wireless Receivers course. The project explored the design and evaluation of an OFDM system tailored for acoustic communication, demonstrating its advantages in handling frequency-selective channels and mitigating inter-symbol interference (ISI).

The transmitter and receiver were carefully designed to ensure efficient data transmission and robust recovery, incorporating features such as preamble-based synchronization, cyclic prefix insertion, and QPSK modulation. To address channel impairments, we implemented channel estimation and phase tracking, enhancing the system's ability to maintain reliable communication under varying conditions.

Our experiments revealed the following key insights:

- **Baseline Performance:** Under good channel conditions, the OFDM system exhibited excellent performance, with received symbols closely matching their corresponding constellation points and minimal errors.
- **Spectral Efficiency:** Reducing the cyclic prefix length to its minimum required value improved spectral efficiency from 0.571 to 0.727 bit/s/Hz without compromising performance. Additionally, increasing the number of sub-carriers further enhanced spectral efficiency by enabling more parallel data transmission, although this required careful management of bandwidth to avoid spectral folding.
- **Phase Tracking:** Continuous phase tracking significantly enhanced system robustness for longer bitstream and dynamic conditions, though its effectiveness was limited under rapid phase changes caused by microphone movement. This limit was treated by increasing the number of training symbols separated by equal duration along the transmission.
- **Real Data Transmission:** Successfully transmitting images demonstrated the system's practical utility, though challenges such as high peak-to-average power ratio (PAPR) required stronger phase tracking and careful signal normalization to preserve data integrity.
- **Robustness Under Different Channel Conditions:** the use of both continuous phase tracking and discrete phase tracking is a good way to correctly estimate a varying channel

The project demonstrated the adaptability and efficiency of OFDM in acoustic environments, particularly in overcoming challenges like multi-path propagation and frequency-selective fading. While the system achieved robust performance under static and mildly dynamic conditions, further improvements, such as adaptive phase tracking, advanced error correction techniques, and dynamic sub-carrier allocation, could enhance its reliability in highly dynamic scenarios.

In conclusion, this project highlights the potential of OFDM as a versatile and effective solution for acoustic communication systems. By addressing key challenges and evaluating performance across a range of conditions, we gained valuable insights into the practical implementation and optimization of OFDM-based systems.